

A Comparative Study of Methods for Early Breast Cancer Detection Using Medical Data

Objective

1. To review and analyze different methods for early detection of breast cancer using medical data.
 2. To build classification models that distinguish between benign and malignant cases.
 3. To compare the performance of these methods using evaluation measures such as accuracy, precision, and recall.
 4. To study how data pre-processing and feature selection affect detection performance.
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Abstract

This project explores different approaches for detecting breast cancer at an early stage using medical data. Early detection plays a crucial role in improving treatment outcomes and survival rates. In this work, several classification techniques are applied to a breast cancer dataset to separate cases into benign and malignant categories. Their performance is then evaluated and compared using standard metrics, including accuracy, precision, and recall. The study also examines how pre-processing steps and the choice of input features influence the results. The findings highlight which method performs best for this task and offer insight into how data-driven techniques can support doctors in making more informed diagnostic decisions.

Tools / Technologies / Database Used

- **Programming language:** Python
 - **Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn
 - **Techniques:** Decision Tree, Random Forest, Support Vector Machine (SVM)
 - **Dataset:** Wisconsin Breast Cancer Dataset
 - **Development environment:** Jupyter Notebook / VS Code
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Expected Outcomes

- A working system to classify breast cancer cases as benign or malignant.
 - Comparison of multiple detection methods.
 - Identification of the best-performing method.
 - Insight into how pre-processing and feature selection affect results.
 - Contribution to research on data-based breast cancer detection.
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